METRICS (How good model is?)

Not used for training, only for evaluation

➢ Accuracy Metrics

- **Accuracy** → % correct predictions
- **BinaryAccuracy** → Accuracy for yes/no
- **CategoricalAccuracy** → Accuracy for one-hot labels
- **SparseCategoricalAccuracy** → Accuracy for integer labels
- **TopKCategoricalAccuracy** → Correct if answer is in top K predictions
- **SparseTopKCategoricalAccuracy** → Same but for integer labels

➢ Probabilistic Metrics

- **BinaryCrossentropy** → Error for binary classification
- **CategoricalCrossentropy** → Error for multi-class
- **SparseCategoricalCrossentropy** → Error for integer labels
- **KLDivergence** → Difference between probability distributions
- **Poisson** → Error for count prediction

➢ Regression Metrics

- **MeanSquaredError** → Average squared error
- **RootMeanSquaredError** → Square root of MSE (more interpretable)
- **MeanAbsoluteError** → Average absolute error
- **MeanAbsolutePercentageError** → % error
- **MeanSquaredLogarithmicError** → Relative error measure
- **CosineSimilarity** → Measures similarity direction
- **LogCoshError** → Smooth error metric
- **R2Score** → How well model explains data (goodness of fit)

➢ Classification (TP/FP based)

- **AUC** → Measures model's ability to separate classes
- **Precision** → Out of predicted positives, how many are correct
- **Recall** → Out of actual positives, how many found
- **TruePositives** → Correct positive predictions
- **TrueNegatives** → Correct negative predictions
- **FalsePositives** → Wrong positive predictions
- **FalseNegatives** → Missed positives
- **PrecisionAtRecall** → Precision at a given recall level
- **RecallAtPrecision** → Recall at a given precision level
- **SensitivityAtSpecificity** → Recall at fixed specificity
- **SpecificityAtSensitivity** → Specificity at fixed recall
- **F1Score** → Balance of precision & recall
- **FBetaScore** → Weighted F1 score
- **PearsonCorrelation** → Linear relationship measure
- **ConcordanceCorrelation** → Agreement between predictions & actual

➢ Image Segmentation Metrics

- **IoU** → Overlap between predicted & actual area
- **BinaryIoU** → IoU for binary images
- **OneHotIoU** → IoU for one-hot labels
- **OneHotMeanIoU** → Average IoU across classes
- **MeanIoU** → Overall IoU score

➢ Hinge Metrics

- **Hinge** → Margin-based accuracy
- **SquaredHinge** → Penalizes errors more
- **CategoricalHinge** → Multi-class margin metric

➢ Wrappers

- **MeanMetricWrapper** → Custom metric wrapper
- **Mean** → Average value
- **Sum** → Total value